

MACHINE LEARNING FOR SIGNAL PROCESSING

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<http://leap.ee.iisc.ac.in/sriram/teaching/MLSP21/>



Overview

- ❖ What are the typical real-world signals
- ❖ What is learning
- ❖ Why should we attempt learning of such signals
- ❖ Roadmap of the course

Real World Signals

- ❖ Signal in general is a function $f : X \rightarrow V$
- ❖ Real World Signals
 - ✓ which we see everyday everywhere
 - ✓ Text, Speech, Image, Videos...
 - ✓ DNA sequence, financial data, weather parameters, neural spike train...
 - ✓ Belonging to / generated by certain category of events.

Real World Signals

- ❖ Types of signals- Continuous and Discrete
- ❖ Observations from real world signals
 - ⦿ Information may not be uniform.
 - ⦿ Cannot be modeled deterministically.
 - ⦿ Affected by noise, sensing equipments.
 - ⦿ Missing or hidden variables.

Real World Signals - Examples

- ❖ Text data
 - ✓ Discrete sequence of items

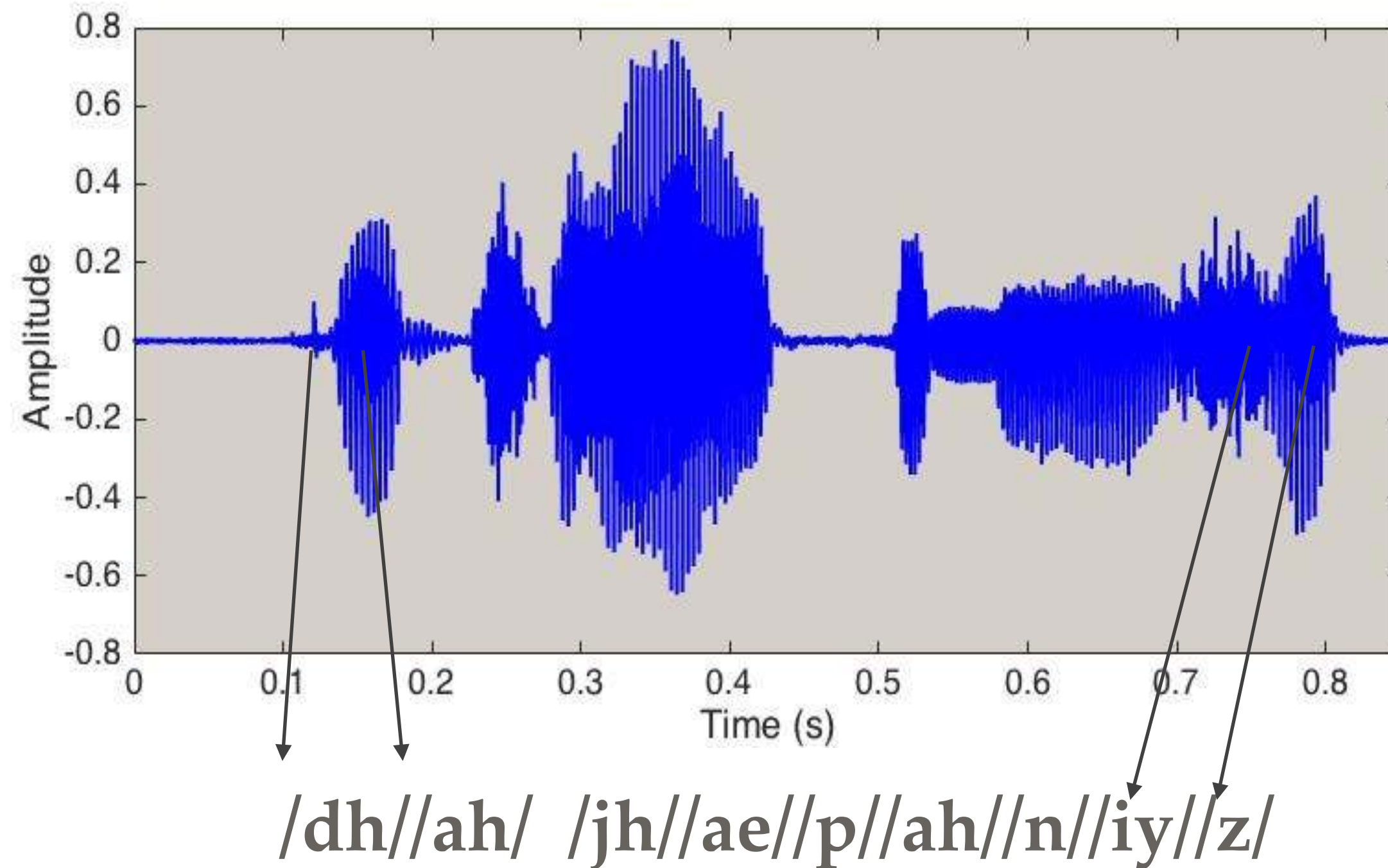
In the last 29 years, sir has never ever said 'well played' to me because he thought I would get complacent and I would stop working hard.

Items - [In] [the] [last] [29] [years]

- ✓ Some items carry more **importance** than others.

Real World Signals - Examples

❖ Speech data



- ✓ Phonetic units - underlying hidden variables.

Real World Signals - Examples

- ❖ Images



- ❖ Measurement artifacts - noise.

Patterns in Real World Signals

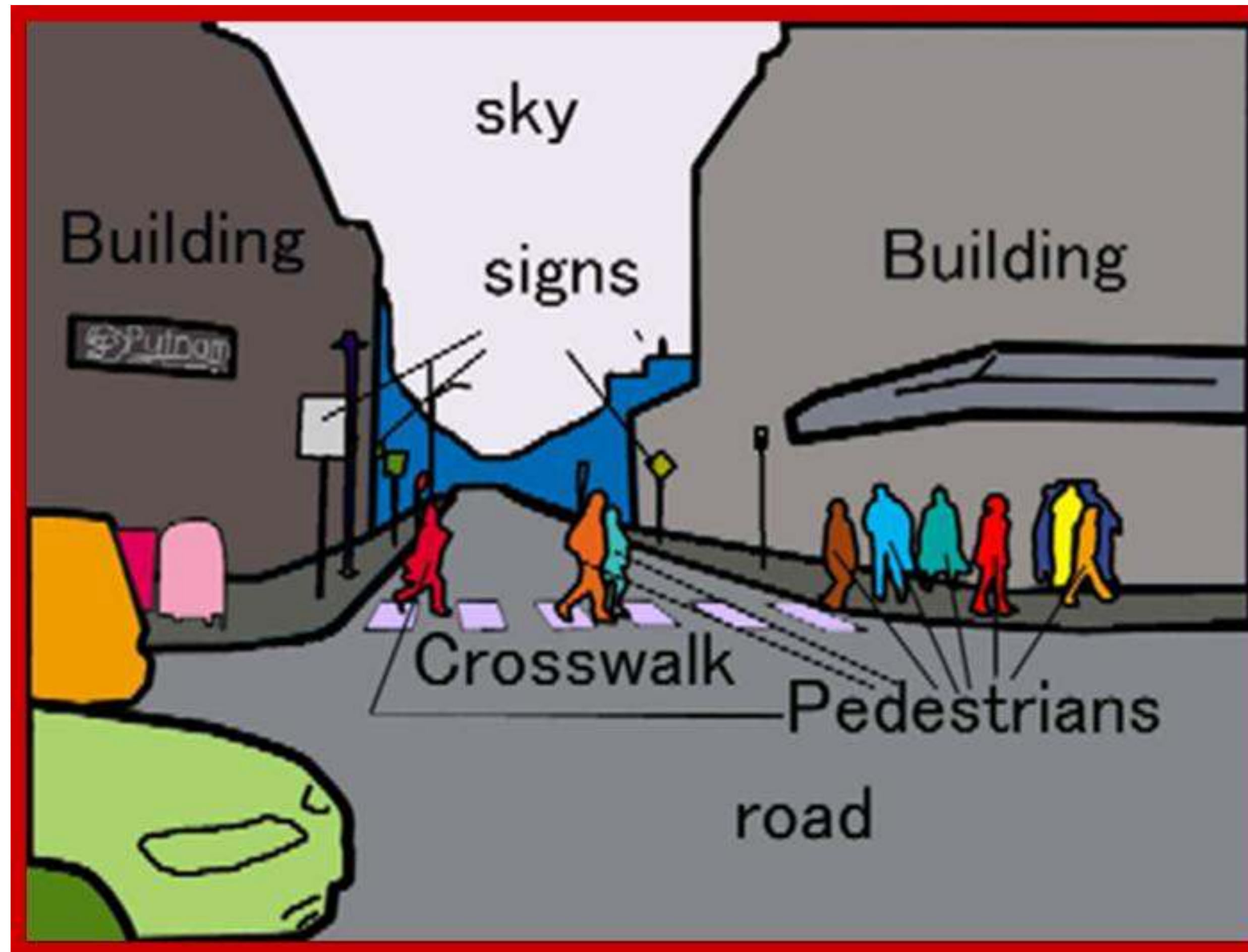
- ❖ Patterns in real world signals
 - ✓ Caused by various generation processes in the real-world signals.
 - ✓ Hidden from the observation.
 - ✓ Value patterns and geometric patterns.
 - ✓ May be hierarchical in nature.
 - ✓ Manifested as pure patterns or transformed / distorted versions.

What is Learning

- ❖ Learning
 - ✓ Process of describing or uncovering the pattern.
 - ✓ Understanding the physical process of generation.
 - ✓ Generalization for prediction, classification, decision making.
 - ✓ Using the data to learn the underlying pattern.
- ❖ Humans are **fundamentally trained** to learn and recognize patterns.

What is Learning

Object
Recognition



What is Learning

Facial Identification



Topic Summarization

The Karnataka government is planning to start an aviation school to help students from lower economic and rural backgrounds become pilots.

Machine Learning

- ❖ Machine Learning
 - ✓ Automatic discovery of patterns.
 - ✓ Motivated by human capabilities to process real world signals.
 - ✓ Mimicking / Extending / Replacing human functions.
 - ✓ Branch of artificial intelligence.
 - ✓ Classification and Regression.

Machine Learning - Examples

Domain Identification - Blog v / s Chat ?

“I tried these Butterscotch Muffins today and they turned out so good. I had half the pack of butterscotch chips that I bought long back so wanted to use it up.”

"Hey, it's Geoff from yesterday. How's it going?Hi there. Don't wanna bother you long, but you saw this video?"

Machine Learning - Examples

Did a Human or Machine write this ?

“A shallow magnitude 4.7 earthquake was reported Monday morning five miles from Westwood, California, according to the U.S. Geological Survey. The temblor occurred at 6:25 AM, Pacific time at a depth of 5.0 miles.”

“Kitty couldn't fall asleep for a long time. Her nerves were strained as two tight strings, and even a glass of hot wine, that Vronsky made her drink, did not help her. Lying in bed she kept going over and over that monstrous scene at the meadow.”

Machine Learning - Examples

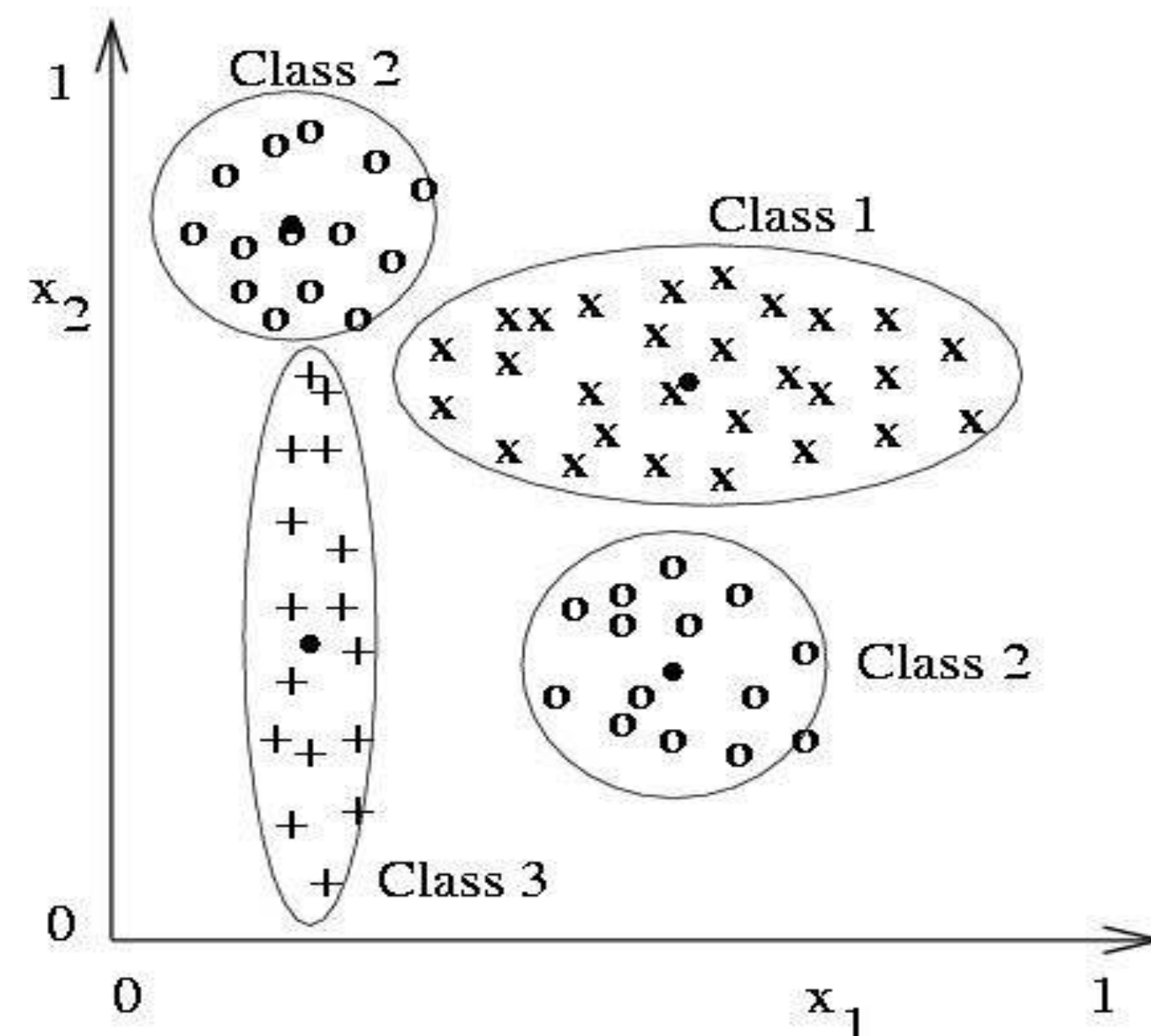
Speech Recognition



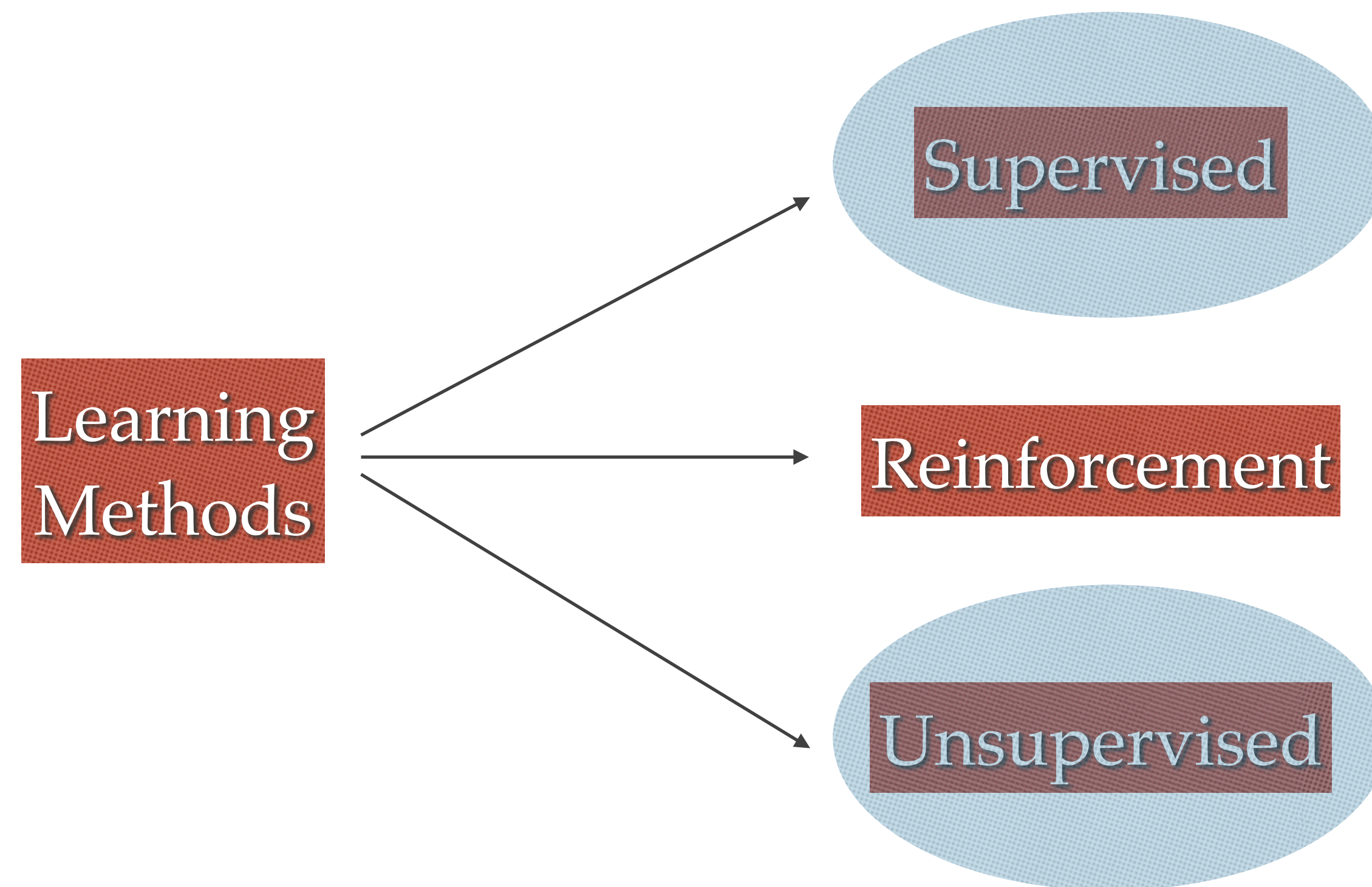
Sound Synthesis

Machine Learning

- ❖ Traditional approaches to Machine Learning
 - ✓ Rule and heuristic based methodologies
 - ✓ Using small amounts of data.
- ❖ Recently, most problems are addressed as statistical pattern recognition problem with big data.



Types of Learning



Unsupervised Learning

- ❖ Data is presented without associated output targets
 - ✓ Extracting structure from the data.
 - ✓ Examples like clustering and segmentation.
 - ✓ Concise description of the data - dimensionality reduction methods.

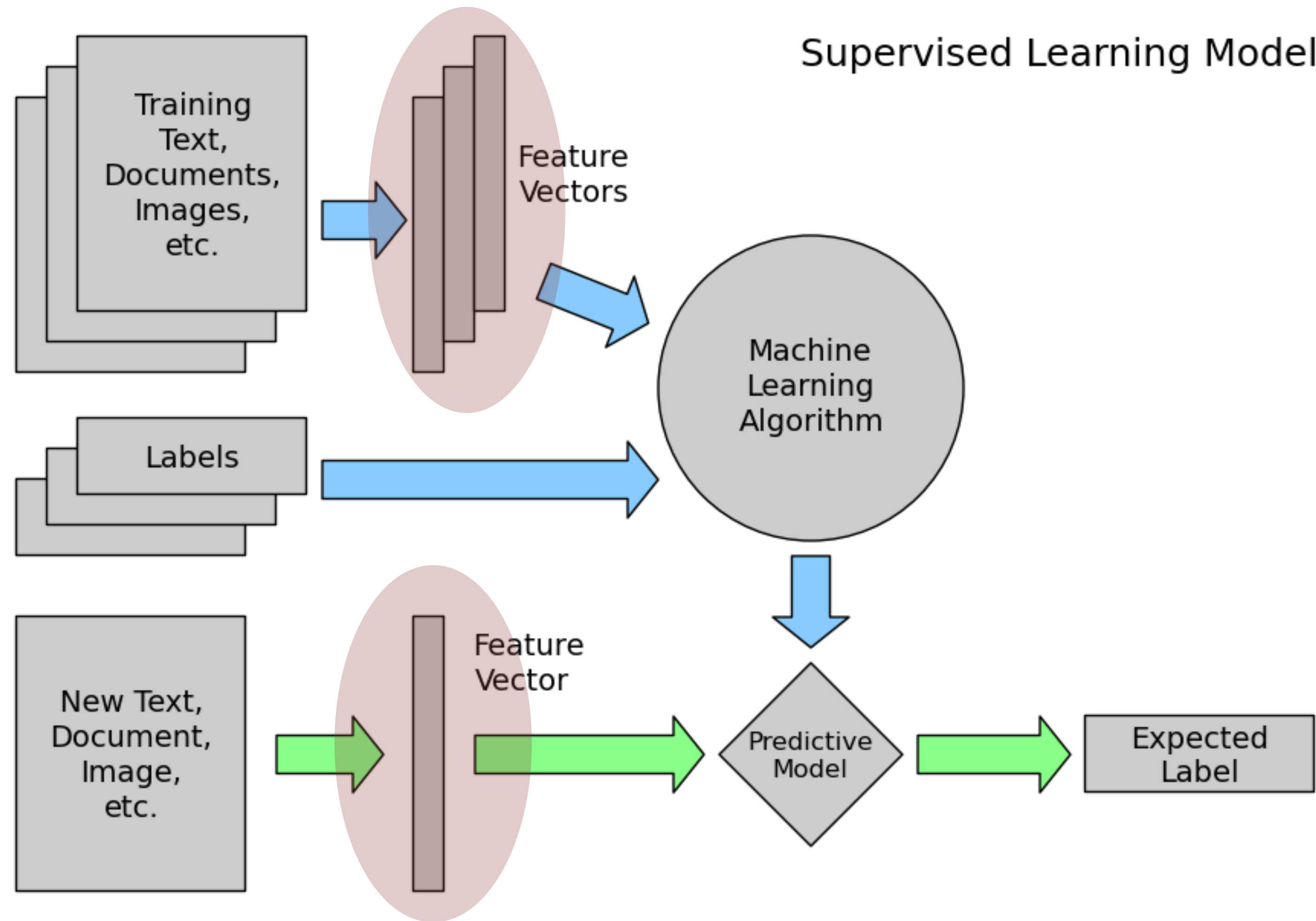
Reinforcement Learning

- ❖ Dynamic environment resulting in triplets - state / action / reward.
 - ✓ No optimal action for a given state
 - ✓ Algorithm has to learn actions in a way such the expected reward is maximized over time.
 - ✓ May also involve minimizing punishment.
 - ✓ Reward / punishment could be delayed - learning based on past actions.

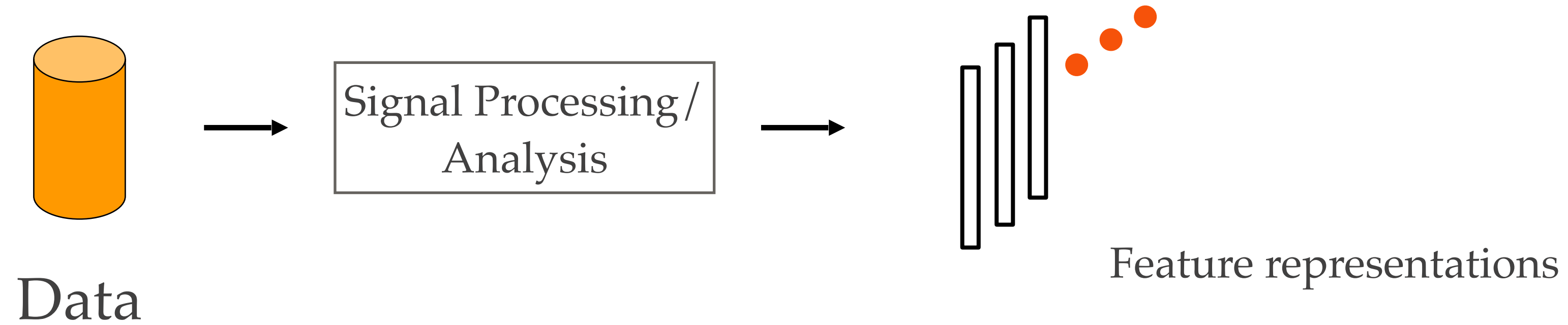
Supervised Learning

- ❖ Training data is provided with along with target values (ground truth).
 - ✓ Goal - to learn the mapping function from data to targets.
 - ✓ Use the mapping function to predict unseen/ test data samples.
- ❖ Two types based on the structure of the labels.
 - ✓ Classification - discrete number of classes or categories.
 - ✓ Regression - continuous output variables.

Supervised Learning

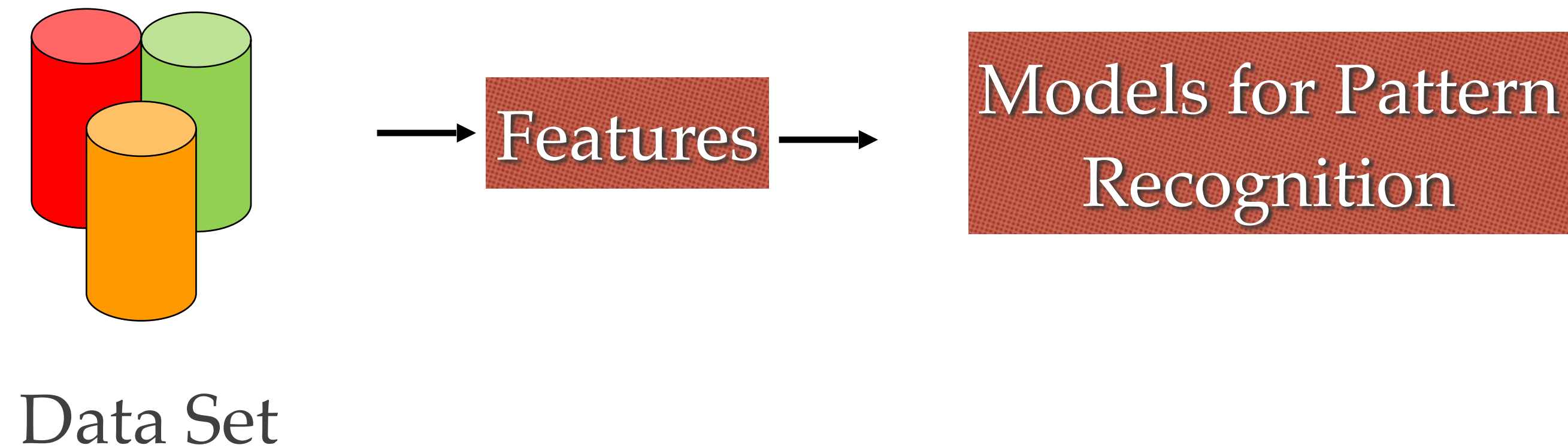


Course Roadmap



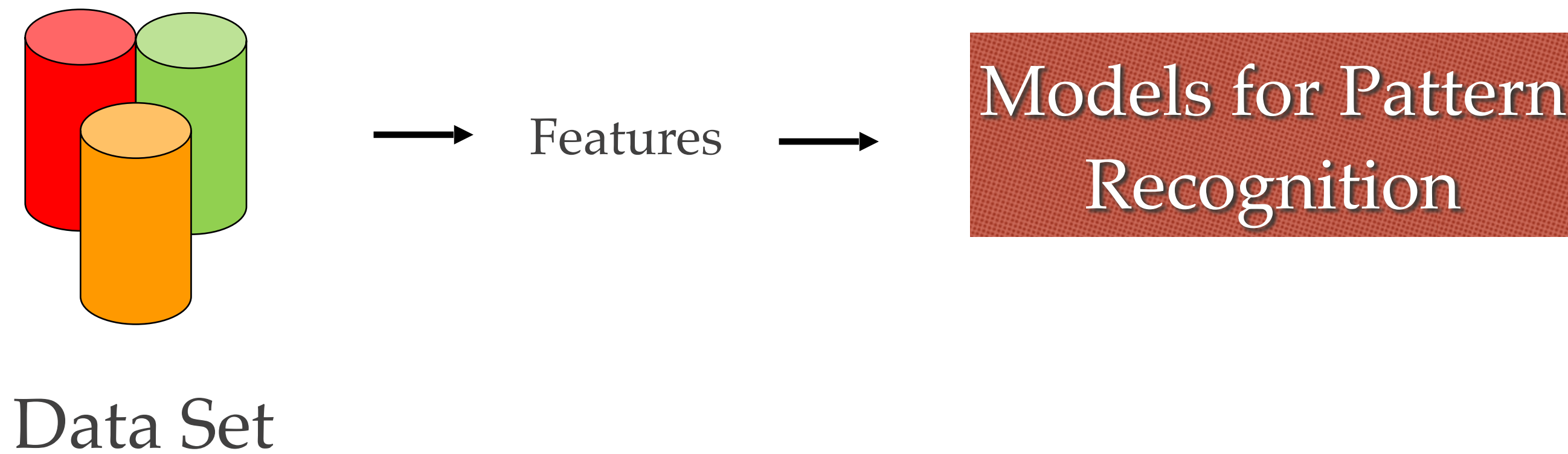
- ❖ Feature Extraction from Text, Speech, Image / Video signals.

Course Roadmap



- ❖ Between features and pattern recognition
 - ✓ Feature selection, dimensionality reduction.
 - ✓ Representation learning.

Course Roadmap



- ❖ Modeling the generation of data
 - ✓ Gaussian, Mixture Gaussian, Hidden Markov Models etc.
- ❖ Modeling the separation of data
 - ✓ Support Vector Machines, Deep Neural Networks etc.

Course Structure (Rough Schedule)

- ❖ Introduction to real world signals - text, speech, image, video.
- ❖ Feature extraction and dimensionality reduction - principal components, linear discriminants.
- ❖ Decision theory for pattern recognition, ML and MAP methods, Bias-variance trade-off, model assessment, cross-validation, estimating generalization error.
- ❖ Generative modeling and density estimation - Gaussian and mixture Gaussian models, kernel density estimators, hidden Markov models. Expectation Maximization.
- ❖ Linear regression and kernel methods. Regularization methods.
- ❖ Discriminative modeling - support vector machines, decision trees and random forest classifiers, bagging and boosting.
- ❖ Neural networks: gradient descent optimization and back propagation, regularization in neural networks, dropout. normalization methods.
- ❖ Introduction to deep learning - feedforward, convolutional and recurrent networks, practical considerations in deep learning.
- ❖ Introduction to graphical models - directed and undirected graphs, belief propagation.

Housekeeping

Requisite	❖ Must
	✓ Probability / Random process / Stochastic Models
	✓ Linear Algebra / Matrix Analysis
	❖ Preferred
Grading	✓ Coding in Python
	❖ Assignments - Theory + Implementation (30%)
	❖ Mid-terms (15%)
	❖ Project (20%)
	❖ Finals (35%)

Housekeeping

Project and Coding Assignments

- ❖ Coding and submissions
 - ✓ Preferred Language - Python.
- ❖ In class demos and example recipes in python.

Resources

- ❖ Textbooks -
 - ❖ PRML (Bishop), NN (Bishop).
 - ❖ Deep Learning (Goodfellow)
- ❖ Online resources (papers and other textbooks listed in webpage).

Course Webpage

www.leap.ee.iisc.ac.in/sriram/teaching/MLSP21



Dates of Various Rituals

- ❖ 6 Assignments spread over 3 months (roughly one assignment every two weeks).
- ❖ April middle - Midterm
- ❖ April 4th week - project topic and team finalization and proposal submission. [1 and 2 person teams].
- ❖ May 3rd week - Project MidTerm Presentations.
- ❖ May last week - Final Exam

Content Delivery

Theory
and Mathematical
Foundation

Intuition and
Analysis

Implementation
and Understanding

Lecture and
Beyond

- ❖ Teaching Assistant - More may be added
- ❖ Course time
- ❖ Industry research lectures (1-2)

THANK YOU

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